The Data-Centre Energy-Control Claim Qualification Benchmark

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Abstract

Average cost can favor a data-centre controller that sometimes fails its service requirement. Results can also change with the training seed, end-of-episode accounting, or the information available to an offline optimizer. We present the Data-Centre Energy-Control Claim Qualification Benchmark, a reanalysis that checks service compliance before comparing costs. PPO, SAC, TD3, and A2C were trained in 12 energy-source configurations with five predeclared seeds and evaluated on the same 103 non-overlapping weeks from 2024–2025. Training and normalization use only 2020–2023 data. The evaluation weeks were excluded from training and have already been used in this retrospective reanalysis. Controllers and hindsight references share the same end-of-episode accounting. An algorithm–configuration comparison qualifies for the primary cost analysis only if all five trained seeds meet the predeclared service criterion.

The primary study contains 240 learned policies and 48 baseline records. Of the learned policies, 136 meet the service criterion individually. Only 4 of 48 algorithm–configuration combinations meet it for all five seeds. The planned cost comparison against the rule-based baseline requires all 48 combinations to qualify. With this requirement unmet, a superiority claim adjusted for multiple comparisons cannot be made. The three other baseline comparisons have the same limitation. A separate accounting check covered 209 supplemental records over 103 episodes of 168 hours each: 3,616,536 record-hours. The maximum absolute discrepancy was zero in all 27 checked quantities. The secondary foresight and objective-weight comparisons include policies with service violations, which limits their relevance when choosing a service-compliant controller.

In a follow-up experiment, we replaced persistence price summaries with weekly seasonal-naive summaries while leaving a threshold policy unchanged. Both versions met the service requirement in all 12 configurations and 103 weeks. Depending on the configuration, the new input reduced end-of-episode-adjusted cost by USD 37,691–106,254 over the evaluation period and captured 5.075%–9.162% of the gap between persistence and hindsight. The opportunity-weighted figure is 7.414%. Dispatch followed the same fixed thresholds in both versions. The dollar reductions sum to USD 707,694.44 across 12 alternative configurations; the configuration-specific amounts describe the saving associated with each simulated design.

Two further follow-up experiments examined recurrent control and training duration. They evaluated 100 policies on the same 103 weeks; 50 met the service criterion. RecurrentPPO failed the all-five-seed screen in each of four configurations. All six planned cost-comparison sets were incomplete after the service check. Of 12 predeclared training-budget comparisons, one recovered compliance, nine remained non-compliant, and two lost compliance. Differences in network size and training recipe limit the interpretation of recurrence, while the selected interaction budgets measure sensitivity to training duration.

The benchmark makes the conditions behind a controller cost claim explicit and tests whether the claim survives them. Its findings apply to the simulated setting, tested controllers, seeds, and budgets. A separate prospective evaluation using the reserved 2026 period remains pending.

1. Introduction

Data centres can draw on grid electricity, local renewables, dispatchable generation, and storage while adjusting cooling and deferrable computing work. Decisions in one hour affect later costs and service. Reinforcement learning (RL), rule-based control, and model-predictive control (MPC) have all been studied for this setting. A lower average simulated cost is not sufficient evidence of a better controller: the comparison must also account for what the controller knew at decision time, whether it met service requirements, how results vary across trained seeds, and what remains at the end of an episode.

Averaging compliant and non-compliant policies can hide failures. An offline optimizer with advance knowledge of the whole episode provides a reference for remaining cost opportunity in the model. Interpreting that reference requires accounting for its information advantage. Even a small error in accounting for unfinished work or battery charge at the final hour can change a close comparison.

We ask whether a cost conclusion still holds after these checks. Every predeclared comparison must be present; each required seed must meet the service criterion; the statistical procedure must support the comparison; and controllers and references must share the same environment and end-of-episode rules. A favorable descriptive ranking does not pass if these conditions are unmet.

1.1 Research questions and follow-up experiments

We rebuilt the comparison with training-only preprocessing, observable system state, common end-of-episode accounting, and an optimizer checked against the simulation equations. These checks establish a common basis for evaluating service and cost across controllers.

Only four of 48 algorithm–configuration combinations passed the five-seed service screen. We then examined three possible explanations for the observed performance. First, we changed the price information supplied to a fixed threshold policy while keeping its dispatch rules unchanged. Second, we compared a recurrent PPO recipe with a feedforward companion in storage-rich configurations, accounting for their different network sizes in the interpretation. Third, we continued training from the one-million-interaction checkpoints toward 2M and 5M to test sensitivity to training duration. A separate replay checked accounting for additional foresight, objective-weight, and storage-size results.

All three follow-up experiments were designed after the primary analysis and are exploratory. The recurrent and budget comparisons were specified before their own held-out evaluation. The price-input experiment is reported descriptively. None of these later results can make the primary comparison complete. A separate 2026 period is reserved for prospective confirmation.

The study addresses five questions:

1. Which algorithm–configuration combinations meet the service requirement across all five trained seeds?
2. How far are eligible online controllers from a hindsight optimizer using the same environment model?
3. What do the secondary foresight, objective-weight, and storage-size comparisons show after full accounting checks?
4. How much can causal weekly price information change one fixed online threshold policy without changing its physical rules or thresholds?
5. Did recurrent control or longer training change which policies met the service requirement?

The contribution is an evaluation procedure for determining which controller cost claims withstand service requirements, seed variability, consistent accounting, and statistical correction. Figure 1 summarizes that procedure.

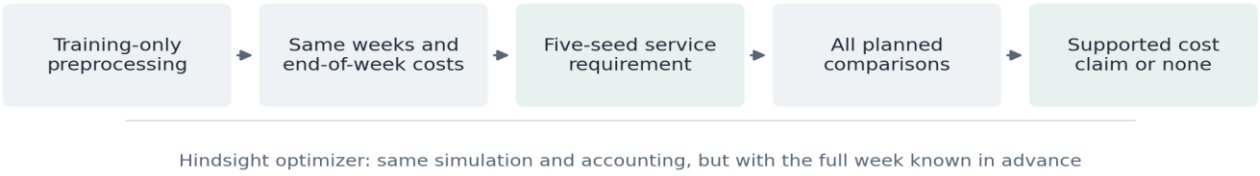


Figure 1. A primary cost claim requires training-only preprocessing, common simulation and end-of-episode accounting, service compliance for all five seeds, and a complete family of comparisons. The hindsight optimizer has perfect information about the full episode and provides an offline reference for the same model.

2. Related work

2.1 Data-centre control, storage, and forecasts

Learned and optimization-based control for data centres is well established, and the broader RL evidence has been synthesized in a recent systematic review (Kahil et al., 2025). Lazic et al. developed genuine learned-model, constrained receding-horizon control for data-centre cooling (Lazic et al., 2018). Model-free RL has also been evaluated for data-centre energy saving (Mahbod et al., 2022). DeepPM combined history-sensitive RL with UPS battery and thermal flexibility (Shao, Islam, and Ren, 2020), while hierarchical MPC has been studied for hybrid energy-storage dispatch in internet data centres (Wang et al., 2023). Recurrent and latent-state policies have been examined for partially observable building and data-centre HVAC control (Biemann et al., 2021) and for response to real-time prices (Biemann et al., 2023).

Recent benchmark and integrated-control work broadens the scope further. DC-CFR combines workload shifting, cooling, and battery control across regions (Sarkar et al., 2024b); DCRL-Green offers a configurable digital-twin benchmark (Sarkar et al., 2024a); and SustainDC includes workload, cooling, battery, forecasts, and cost/carbon/water objectives (Naug et al., 2024). Integrated rolling optimization with workloads, renewables, batteries, and forecasts has also been published (Yang et al., 2025), while decision-focused forecast-to-control learning continues to develop (Yang et al., 2026). These studies

preclude a broad novelty claim based only on combining RL, recurrence, forecasts, renewable sources, and storage.

2.2 Evaluation reliability in reinforcement learning

RL comparisons depend on implementation choices, hyperparameters, random seeds, and reporting practices (Henderson et al., 2018). The number of seeds affects statistical uncertainty (Colas et al., 2018). When few runs are available, confidence intervals and robust summaries help show that uncertainty (Agarwal et al., 2021).

Our study requires service compliance before the primary algorithm comparison. The five seeds represent five independently trained policies. Repeating the same evaluation weeks measures their behavior under common conditions; the independent training sample remains five policies per algorithm and configuration. Estimating operational failure probability would require a deployment study.

2.3 Positioning of the present study

Prior work already establishes the main controller ingredients. Our comparison differs in how it tests their reported advantages: training-derived preprocessing is reused without refitting; all controllers face the same held-out weeks; the hindsight reference follows the same environment equations; end-of-episode costs are included; and every required seed must meet the service criterion before a primary cost claim is tested.

3. Methods

3.1 Data periods and simulated setting

Training and preprocessing use data from 1 January 2020 through 31 December 2023. Evaluation uses 103 non-overlapping, 168-hour weeks from 2024–2025, excluded from training. Their previous use in this retrospective reanalysis is distinct from the prospective evaluation reserved for 2026. Every controller faces the same weeks in the same order. Evaluation reuses normalization statistics calculated from the training period.

The study covers 12 energy-source configurations, from grid-only operation to combinations of solar, wind, dispatchable gas generation, and battery storage. Battery capacity and power are fixed in the main comparison. The controller observes battery charge, deferred-work backlog, whether a service failure has occurred, time remaining in the week, current external conditions, and forecast summaries.

The predeclared objective includes cost, carbon, water, service, and end-of-episode adjustments. Controllers use the same accounting rules. Adjustments for unfinished work and battery charge prevent a controller from appearing cheaper merely by ending the week with a backlog or depleted storage.

3.2 Primary controller population

The primary learned population comprises PPO, SAC, TD3, and A2C trained across 12 configurations and five seeds (42, 123, 456, 789, and 1024), for 240 policies. Four baselines—a no-intervention policy, a greedy policy, a rule-based policy, and a fixed-threshold policy—are evaluated in every configuration, producing 48 baseline records. Each record is paired to the same 103 held-out weeks. The primary online controllers use persistence price summaries: the current price supplies the estimates for the next four and 24 hours.

Each seed trajectory in the figures represents one independently trained policy. Each algorithm has five such trajectories per configuration.

Two offline hindsight optimizers provide references: one minimizes cost without the service constraint, and the other enforces service compliance. Both use the same environment equations and end-of-episode accounting as the controllers. Each optimizer receives the full 168-hour week before solving its dispatch schedule. This perfect information defines the model's hindsight cost reference. The primary online controllers make decisions sequentially using the observations available at each hour. Their costs and service outcomes are compared with this reference to measure the remaining opportunity within the simulation.

3.3 Qualification-first primary analysis

We determine service eligibility by replaying each policy in the environment. An algorithm–configuration combination enters the primary cost comparison only if all five required seeds have zero recorded service violations across the 103 evaluation weeks. Failed policies remain part of the reported population and determine whether the combination qualifies. The modeled violation counter increments when accumulated deferred load, after adding the current hour's deferral, exceeds 0.30×12 times that hour's base demand. The environment then forces the excess into served demand. This is an aggregate backlog constraint; it does not track the age or deadline of each individual job.

The primary analysis specifies 48 comparisons against the rule-based baseline. Under the predeclared study protocol, Holm correction is applied only when the complete family meets the service requirement. Failure of any member limits the family to descriptive reporting. This complete-family eligibility rule is a study-specific requirement. The declared policy population remains fixed, including missing or failed policies.

For eligible comparisons, the predeclared method uses 10,000 bootstrap resamples across trained seeds and paired weeks. A separate fixed random-number seed, 20260803, makes the resampling calculation reproducible. It was specified before evaluation and is distinct from the five seeds used to train policies. Its numerical value has no substantive interpretation. We report two-sided 95% percentile intervals and centered two-sided resampling p-values. The same weeks remain paired across controllers, and the independently trained policy remains the training unit.

Each multiple-testing family is the full set of comparisons covered by one statistical correction. For service-qualified controllers, the primary hindsight gap is the percentage of their aggregate cost that could be removed by the service-compliant hindsight optimizer: $100 \times (\text{controller cost} - \text{hindsight cost}) / \text{controller cost}$. The reported intervals use the predeclared paired resampling procedure.

3.4 Supplemental verification and robustness families

The supplemental population comprises 120 foresight policies, 80 Pareto policies, five 40-MWh policies, and four 40-MWh baselines: 209 records in total. The accounting check replayed every record for all 103 weeks and all 168 hours per week. It checked 13 hourly and 14 end-of-episode quantities against predeclared tolerances and rejected missing, duplicate, non-finite, inconsistent, or incomplete values.

In the foresight experiment, PPO and SAC receive exact future price averages for the next four and 24 hours at each decision. This information differs from the full-week information supplied to the hindsight optimizers and from the historical inputs used in the seasonal-naive experiment. Foresight and objective-weight comparisons use their own predeclared multiple-testing families and include policies with service violations. Their results are therefore interpreted alongside service compliance. The battery-capacity

comparison covers 20 and 40 MWh. In the 40-MWh PPO group, two of five seeds account for nine service violations across four episodes.

3.5 Causal price-forecast-input ablation

The price-input experiment ran the same deterministic threshold policy across all 12 configurations and 103 evaluation weeks. One version received persistence summaries; the other replaced two price summaries with values based on the corresponding hours of the previous week. Before a full prior week was available, it used persistence. Policy thresholds, simulated physics, weeks, objective, storage settings, and end-of-episode accounting stayed the same.

The policy selects actions using preset thresholds. We refer to its two versions as the persistence threshold policy and the seasonal-naive-price threshold policy. Changing their inputs while keeping their decision rules fixed isolates the effect of the supplied price summaries. The service-compliant hindsight optimizer provides the cost reference.

The aggregate normalized forecast-input effect is

$$\frac{\sum_e C_{\text{persistence},e} - \sum_e C_{\text{seasonal},e}}{\sum_e C_{\text{persistence},e} - \sum_e C_{\text{hindsight},e}}.$$

This quantity measures the proportion of the persistence-to-hindsight cost gap captured by the changed input for this policy. Its denominator is the difference between persistence-policy cost and hindsight cost. The ratio can fall outside zero to one in other outcomes, and its interpretation depends on the tested policy and information source.

3.6 Recurrent-control and training-duration experiments

The recurrent-control experiment evaluated RecurrentPPO in four storage-rich configurations, with five seeds and a target of one million environment interactions. The earlier feedforward PPO used different rollout and minibatch settings, so we also trained 20 feedforward companion policies (four configurations × five seeds). These used the same `n_steps=512`, `batch_size=128`, learning rate, epochs, discount, GAE, clipping, activation, vectorization, evaluation rules, and interaction target, but no recurrent state. The specified 21-observation/4-action networks have approximately 11,465 trainable parameters for companion PPO and 332,873 for RecurrentPPO—a 29.03× size difference. The experiment therefore compares two training recipes; it cannot attribute any difference to memory alone.

The training-duration experiment continued PPO, SAC, and RecurrentPPO in two configurations: grid, solar, wind, gas generation, and battery; and grid, solar, wind, and battery. Thirty algorithm–configuration–seed runs resumed from full-state 1M checkpoints and continued once toward 5M, with 2M and 5M snapshots fixed in advance: 60 additional evaluated policies. All checkpoints, budgets, and architectures were specified before evaluation. The comparison measures sensitivity to training duration within these runs.

Before evaluation, we specified which policies and weeks to use, how to reset recurrent state, how to handle failed policies, and how to calculate and interpret the comparisons. The recurrent experiment has three four-comparison cost families. The training-duration experiment has a primary family of 12 comparisons against 1M, a secondary family of six 5M-versus-2M comparisons, and a secondary family of 18 comparisons against the rule-based baseline. Thus, the six families contain 4, 4, 4, 12, 6, and 18 comparisons. Every member must satisfy the all-five-seed service criterion before the family's Holm correction is applied. We report changes in compliance before comparing costs.

We set a 1% minimum cost difference of interest. In these follow-up experiments, the hindsight gap is expressed relative to hindsight cost: 100 × (controller cost − hindsight cost) / hindsight cost. We considered the gap closed only if all five seeds met the service requirement and its 95% upper confidence bound was at most one percentage point. These round thresholds were chosen before evaluation as study-specific criteria. We also report descriptive sensitivity at half and twice those thresholds. Evaluation used the policies' deterministic prediction interface without changing, retrying, or replacing actions.

All 100 follow-up policies were evaluated together once on the same 103 weeks. Their methods were specified before their own evaluation. These exploratory comparisons use the same retrospective evaluation period as the primary study. The scope of the numerical checks is described in Appendix B.3.

4. Results

4.1 Evidence completeness and reference verification

The primary dataset contains 288 records: 240 learned policies and 48 baselines. A numerical consistency check found the expected records, weeks, and numerical arrays. It found no failures in 29,664 comparisons with the unconstrained hindsight bound, 18,952 comparisons with the service-compliant bound, 1,236 checks of bound ordering, 3,708 checks of optimizer gaps, or 2,472 checks linking optimizer results to environment replay. All 1,236 numerical consistency checks for the revised optimizer passed.

The supplemental accounting check covers 209 records and 3,616,536 record-hours. The maximum absolute discrepancy was zero for all 27 checked quantities. The service-compliant 40-MWh hindsight reference met the service requirement in all 103 weeks, and its service counts agreed with the simulation.

These checks support the numerical consistency of the calculations. Assessing transfer to a real data centre also requires validation of the physical model and its operating assumptions.

4.2 Primary qualification result

Of 240 learned policies, 136 individually meet the service criterion. All 48 baselines do. Table 1 shows the number of compliant seed policies for each of the 48 algorithm–configuration combinations. A combination qualifies only when all five policies have zero recorded service violations across the 103 weeks. Exactly four combinations meet that requirement.

Table 1. Compliant trained policies out of five for every primary algorithm–configuration combination. Bold values identify all-five-seed qualification.

Energy-source configuration	PPO	SAC	TD3	A2C
Grid only	3/5	2/5	1/5	2/5
Grid + solar	3/5	4/5	0/5	3/5
Grid + wind	4/5	4/5	1/5	4/5
Grid + gas	3/5	2/5	1/5	3/5
Grid + solar + wind	3/5	4/5	1/5	3/5
Grid + solar + gas	4/5	3/5	2/5	3/5

Energy-source configuration	PPO	SAC	TD3	A2C
Grid + wind + gas	4/5	**5/5**	1/5	3/5
Grid + solar + battery	4/5	**5/5**	0/5	1/5
Grid + wind + battery	3/5	**5/5**	1/5	4/5
Grid + solar + wind + gas	4/5	4/5	1/5	4/5
Grid + solar + wind + battery	**5/5**	3/5	0/5	3/5
Grid + solar + wind + gas + battery	4/5	3/5	2/5	4/5

Each of the four controller-versus-baseline comparison families has four service-qualified combinations and 44 exclusions. This leaves the primary 48-comparison family against the rule-based baseline incomplete and prevents a Holm-corrected primary superiority claim. Cost rankings are reported descriptively across the declared population.

The four qualified combinations retain a measurable cost gap to the service-compliant hindsight optimizer. Table 2 reports every qualified combination using the recorded, predeclared gap calculation. The gap ranges from 14.302% to 24.815% of controller cost. These estimates describe opportunity within the model under perfect information. A deployable controller's ability to realize that opportunity remains a separate question.

Table 2. Recorded cost gaps for all four service-qualified learned combinations. Percentages use controller cost as the denominator. Confidence intervals are the predeclared two-sided 95% bootstrap intervals; the cost difference is the mean paired weekly difference across seeds and weeks. These descriptive gaps do not establish superiority over a baseline.

Qualified controller and configuration	Cost gap (%)	95% interval (%)	Mean weekly cost gap (USD)
SAC: grid + solar + battery	23.067	20.664–25.605	10,406.71
SAC: grid + wind + battery	24.815	22.141–27.851	9,248.91
SAC: grid + wind + gas	14.302	13.266–15.279	5,334.79
PPO: grid + solar + wind + battery	23.564	20.802–26.696	7,320.70

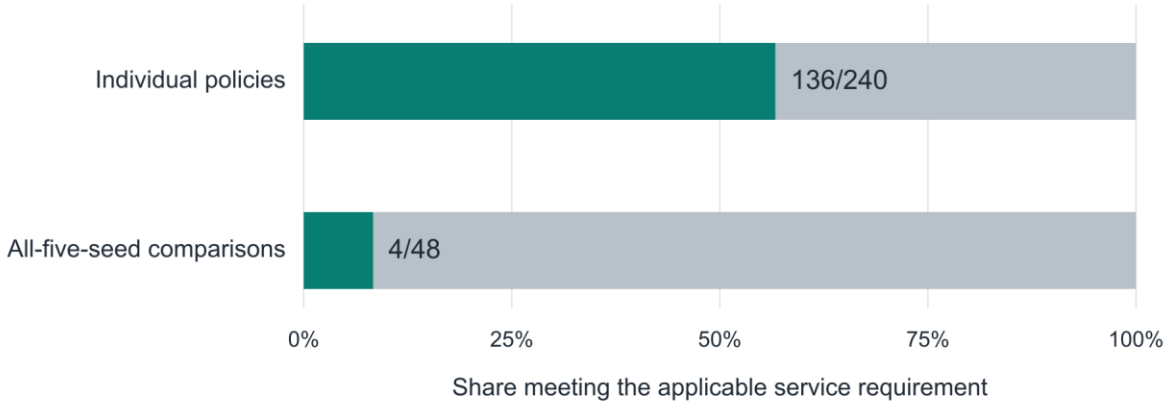


Figure 2. Service compliance for individual trained policies (136/240) and for algorithm–configuration combinations where all five seeds comply (4/48). The first measure concerns individual policies; the second concerns reproducibility across all five training seeds. The incomplete 48-comparison family prevents a Holm-corrected primary cost claim.

Figure 3 shows weekly costs behind the aggregate result. For each algorithm and seed, it plots the percentage difference between that policy's total end-of-episode-adjusted cost across all 12 configurations and the corresponding rule-based baseline total. It includes every algorithm, seed, configuration, and week. Negative values favor the learned policy on cost. The plot includes policies that fail the service requirement, so its cost lines are descriptive.



Figure 3. Weekly cost difference from the rule-based baseline for all four algorithms and five independently trained seeds over 103 weeks. Each line compares weekly cost totals across all 12 configurations for one seed. The plot retains the complete seed and week population, including policies that fail the service requirement.

Figure 4 shows why service outcomes need to accompany cost lines. Service violations differ sharply among seeds of the same algorithm. Some seeds have none; others accumulate violations during different parts of the evaluation period. The totals count violations across 12 simulated configurations. Estimating deployment failure rates would require operational evidence.

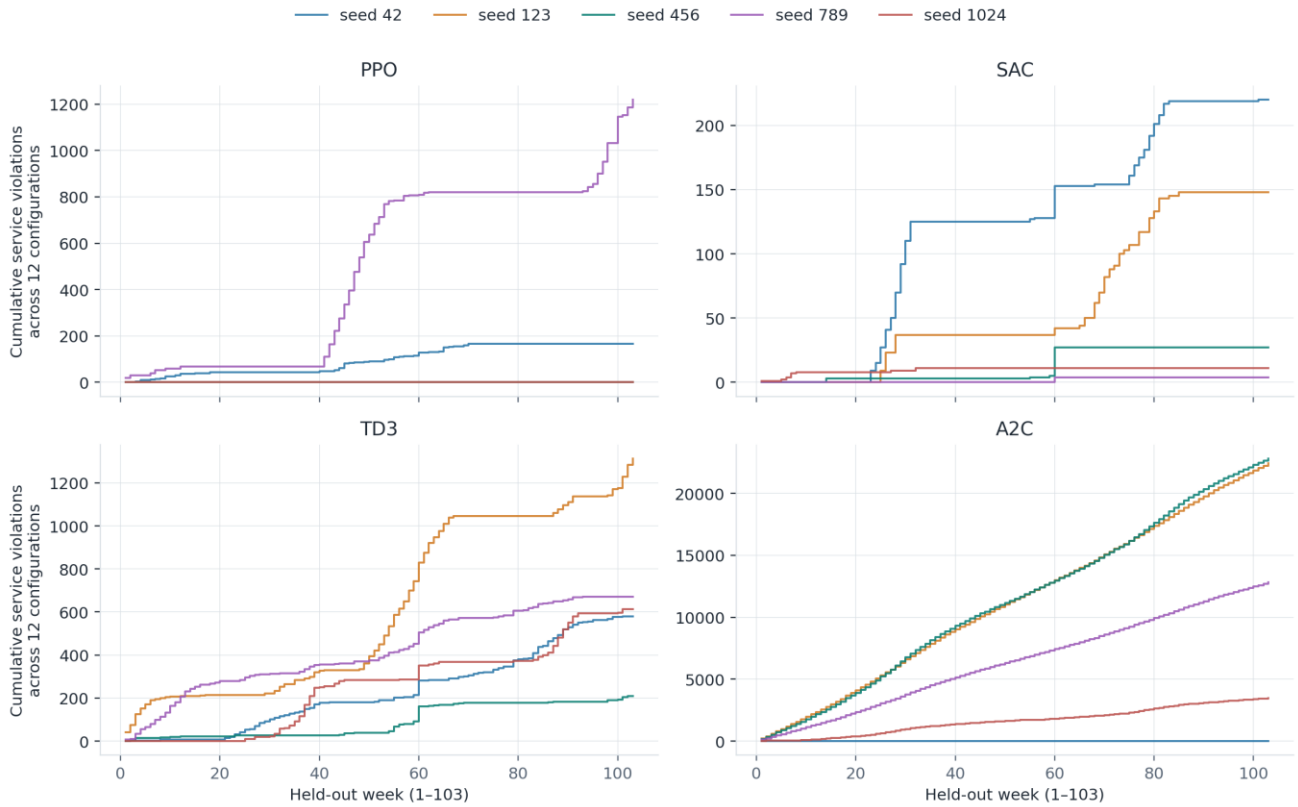


Figure 4. Cumulative service violations across 12 configurations for each algorithm and seed, week by week. Each panel shows all five predeclared seeds. The differing trajectories make the seed sensitivity behind the four qualifying combinations visible.

4.3 Secondary robustness results and their boundary

The foresight analysis contains all 24 planned comparisons and rejects five after Holm correction. Service violations occur in 57 of its 120 policy records. The objective-weight (Pareto) analysis rejects all three planned comparisons and includes 23 service-violating records among 80 policies. These service outcomes limit the use of the secondary comparisons when assessing service-qualified superiority.

For the rule-based baseline, the simple average of four storage-configuration hindsight gaps is 24.277%, combining 22.286%, 23.480%, 24.083%, and 27.257%. Each percentage uses the baseline's cost as its denominator. Pooling costs before calculating the percentage gives 24.042%. The recurrent-control and training-duration experiments express their gaps relative to hindsight cost; comparison across these measures requires keeping the denominators distinct. The battery-capacity results cover 20 and 40 MWh. In the 40-MWh PPO group, two of five seeds account for nine service violations across four weeks. Identifying an optimal battery size would require a broader joint study of capacity, power, and controller behavior.

4.4 Causal forecast-input ablation

We numerically verified both versions of the price-input experiment across 2,472 controller–configuration weeks and 415,296 simulation steps. Replayed costs, service outcomes, and actions matched the stored results, with zero service violations. We report the cost changes descriptively; the scope of the numerical checks is stated in Appendix B.3.

Table 3 summarizes the complete evaluated population.

Table 3. Aggregate end-of-episode-adjusted costs across all 12 alternative configurations and 103 weeks. The summed difference describes this panel of alternatives; configuration-specific reductions are reported in the text.

Quantity	Aggregate terminal-adjusted cost (USD)
Persistence threshold policy	51,659,140.61
Seasonal-naive-price threshold policy	50,951,446.17
Service-compliant hindsight optimizer	42,113,795.28
Seasonal-naive arithmetic panel-sum reduction from persistence	707,694.44
Persistence-to-hindsight opportunity	9,545,345.33
Normalized forecast-input effect	7.414%

Both threshold-policy versions met the service requirement in every week. The improvement relative to the persistence-to-hindsight gap ranges from about 5.075% to 9.162% across the 12 configurations. Each configuration improved in 33 weeks. In the eight configurations without a battery, costs changed in those 33 weeks and never worsened. In the four battery configurations, costs changed in 36 weeks: 33 improved and three worsened. The original check against accidental use of future information sampled five hours, rather than testing every hour; later code inspection and additional tests supported the causal design.

For this fixed threshold policy and these weeks, prior-week price summaries reduced cost. The saving for each simulated design is its configuration-specific reduction of USD 37,691–106,254 over the 103 weeks. The combined USD 707,694.44 describes the arithmetic sum across the 12 alternative designs. The measured effect applies to the tested price summaries, threshold rules, and evaluation period.

4.5 Statistical reproducibility

A numerical reanalysis reproduced the primary methods, tables, figure labels, rankings, multiple-testing decisions, and reported conclusions. With a different NumPy quantile implementation, 29 lower confidence-interval endpoints differed by one to five units in their last decimal place. The differences were at machine precision and changed no interval interpretation, p-value, Holm decision, ranking, or conclusion.

4.6 Exploratory history-conditioned and budget extensions

All 100 recurrent-control and training-duration policies were evaluated on the same 103 weeks, totaling 1,730,400 policy–environment steps. Fifty policies met the service requirement. Figure 5 includes every evaluated policy, including those that failed service checks. The scope of the numerical checks is described in Appendix B.3.

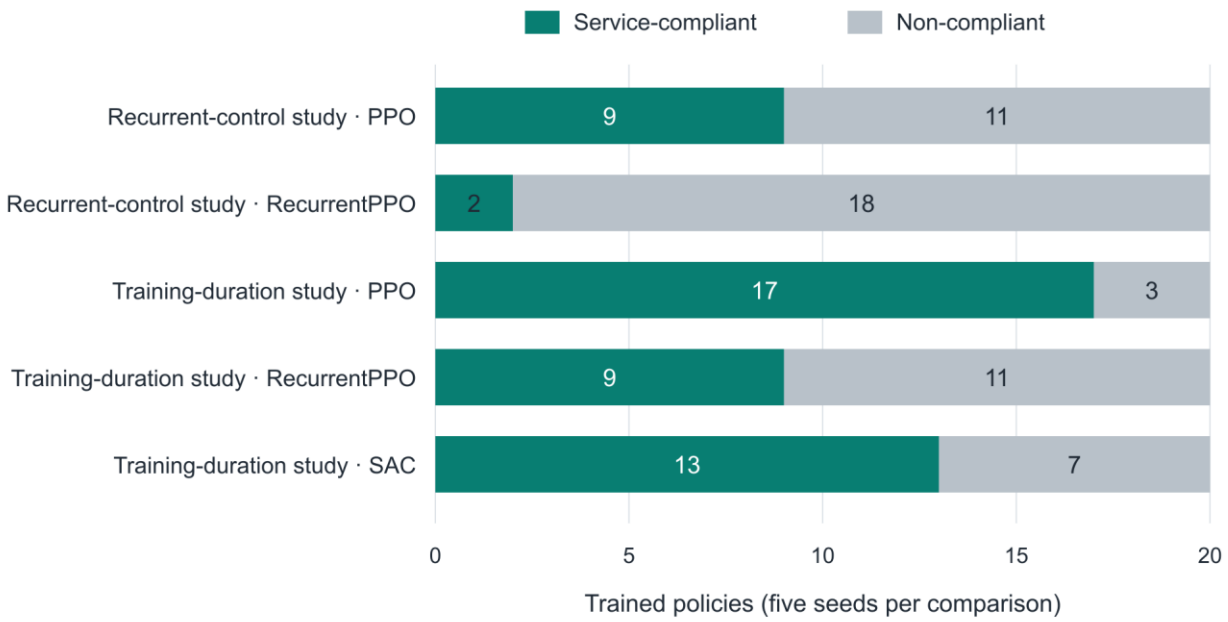


Figure 5. Service compliance in the recurrent-control and training-duration experiments. Each row contains 20 trained policies. A policy qualifies when its recorded service-violation count is zero across the 103 weeks. The bars show service outcomes for all 100 policies.

Table 4. Completeness of each predeclared follow-up cost-comparison family after the all-five-seed service check.

Predeclared cost comparison	Number of comparisons	Complete after service screen?	Holm-adjusted cost claim?
RecurrentPPO versus companion PPO	4	No	No
RecurrentPPO versus rule-based baseline	4	No	No
RecurrentPPO versus earlier PPO	4	No	No
2M/5M versus 1M	12	No	No
5M versus 2M	6	No	No
Training snapshots versus rule-based baseline	18	No	No

RecurrentPPO failed the five-seed service screen in **all four** tested configurations. Only two of its 20 policies met the service requirement individually. Nine of 20 companion feedforward PPO policies did, but no configuration had all five eligible seeds in both groups. We cannot attribute the difference to recurrence alone: the networks differ greatly in size, and other factors may matter.

Among 12 planned comparisons of the 2M and 5M snapshots with 1M, one recovered five-seed service compliance, nine remained non-compliant, and two lost compliance. With all five energy sources available, PPO met the all-five-seed requirement at 2M after failing it at 1M, then failed it again at 5M. At 2M, its cost remained 35.452% above the service-compliant hindsight cost (95% interval: 30.886%–41.928%). The interval's upper bound exceeded the predeclared one-percentage-point criterion for closing the gap. In the grid, solar, wind, and battery configuration, PPO qualified at 1M and lost qualification at both 2M and 5M. The other nine comparisons remained ineligible. Among six secondary 5M-versus-2M

comparisons, one lost compliance and five remained non-compliant. Figure 6 shows every planned primary comparison.

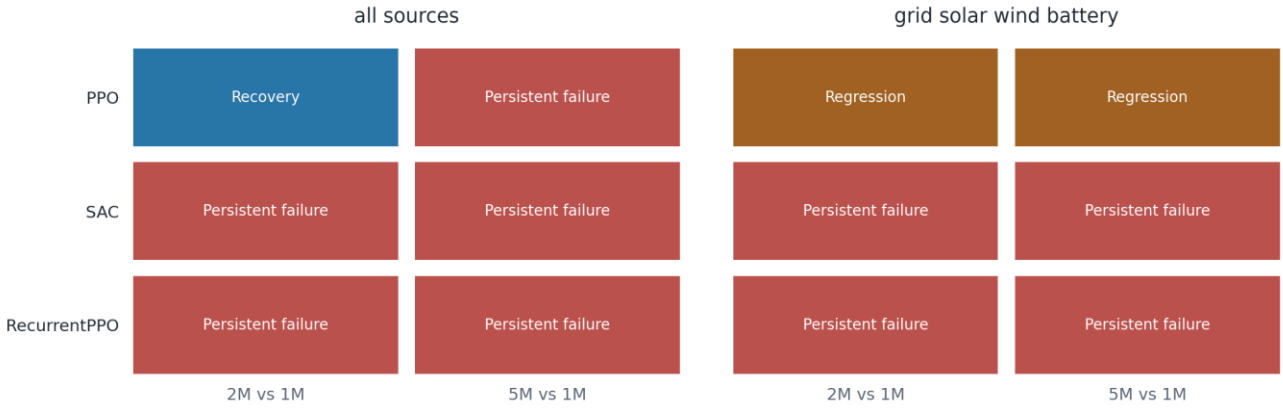


Figure 6. Service-compliance changes from 1M to the preselected 2M and 5M snapshots. Recovery, persistent failure, and regression describe whether all five seeds qualify. The 12 comparisons produce one recovery, nine persistent failures, and two regressions.

Figure 7 shows weekly costs for all five seeds at the 1M, 2M, and 5M snapshots. Faint lines represent individual seeds; the heavier line is their mean. Interpretation of these costs also requires the service outcomes and statistical eligibility reported separately.



Figure 7. Weekly end-of-episode-adjusted cost for every training-duration algorithm, configuration, snapshot, seed, and evaluation week. Columns show PPO, SAC, and RecurrentPPO; rows show the two storage-rich configurations. Colors identify the 1M, 2M, and 5M snapshots. Faint curves show individual seeds and heavy curves their means. All snapshots, seeds, and weeks are shown.

Figure 8 shows service violations for the same 90 trajectories. Longer training affects seeds differently: some have no change, some accumulate fewer violations, and others accumulate more. The predeclared comparisons yield one recovery, nine persistent failures, and two regressions.

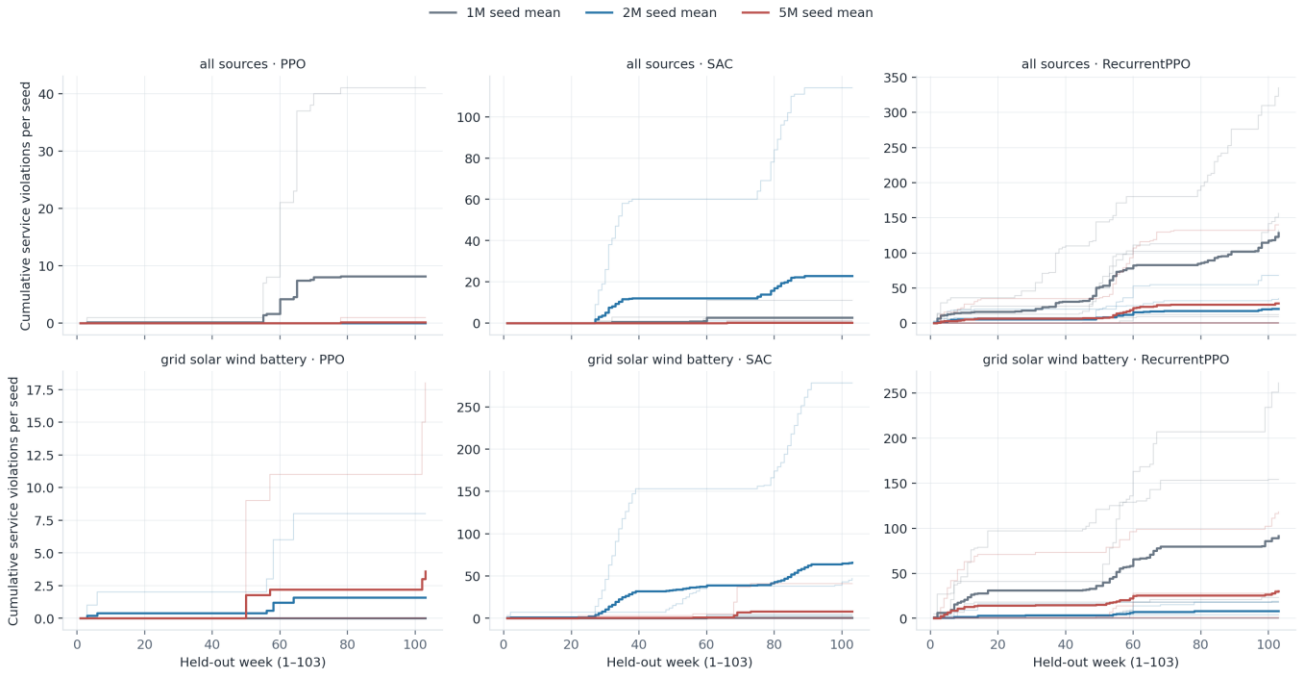


Figure 8. Cumulative service violations by week at the 1M, 2M, and 5M snapshots. Faint curves show individual seeds; heavy curves show the mean across five seeds. The panels use the same order as Figure 7. The differing paths show that service outcomes vary across seeds and training budgets.

All six planned cost-comparison families are incomplete. Their cost estimates and unadjusted p-values remain descriptive. The findings apply to the tested recipes, seeds, configurations, and budgets. The predeclared threshold-sensitivity checks leave service eligibility and the statistical conclusion unchanged.

5. Discussion

The five research questions separate service reliability, remaining cost opportunity, information, controller design, and training duration. Taken together, their answers show where the tested controllers qualify for a cost claim and where the evidence remains insufficient. The distinction matters because a controller can appear economical while its service outcomes vary substantially across independently trained policies.

5.1 Service compliance across independently trained policies

The first question has a direct answer: four of 48 algorithm–configuration combinations met the service requirement in all five seeds. SAC qualified with grid, solar, and battery; grid, wind, and battery; and grid, wind, and gas. PPO qualified with grid, solar, wind, and battery. Table 1 shows all 48 combinations, including those with four compliant seeds and those with none. Across individual policies, 136 of 240 qualified, compared with all 48 baseline records.

This pattern shows why an average over seeds can conceal a decision-relevant difference. A configuration with four compliant policies and one failing policy remains sensitive to which trained policy is used. The all-five-seed requirement makes that sensitivity visible before costs are compared. The 44 combinations that failed the screen leave the planned 48-comparison family incomplete, preventing the predeclared Holm-corrected claim that a learned controller outperforms the rule-based baseline across the comparison set. The four qualifying combinations remain specific candidates for further study. Establishing their

operational reliability would require evidence beyond five training seeds and a simulated evaluation period.

5.2 Remaining cost opportunity in service-qualified controllers

The second question concerns the cost left above a service-compliant optimizer using the same physical and accounting model. For all four qualified learned combinations, the recorded gap was positive: 23.067% for SAC with grid, solar, and battery; 24.815% for SAC with grid, wind, and battery; 14.302% for SAC with grid, wind, and gas; and 23.564% for PPO with grid, solar, wind, and battery. The corresponding 95% intervals in Table 2 remain above zero. These percentages express the gap as a share of each controller's cost.

The common model makes this a meaningful comparison of schedules and accounting. The optimizer's advance knowledge of the entire week also gives it an information advantage. Consequently, the measured gap combines the opportunity associated with perfect information and the differences in dispatch decisions. Separating those contributions requires further experiments with matched information and a deployable optimization policy. The gap provides a reference against which such experiments can be assessed; translating it into a facility saving requires practical forecasting, dispatch, and service validation.

5.3 Foresight and objective weights under service constraints

The third question asks what survives the additional information, objective-weight, and battery-capacity comparisons. Exact future price averages changed some cost comparisons: five of the 24 foresight tests rejected their null hypotheses after Holm correction. The objective-weight analysis rejected all three planned targeted-objective contrasts. These findings show that the additional inputs or objective choices affected the tested comparisons. Their relevance to choosing a compliant controller is limited by the service violations present in 57 of 120 foresight policies and 23 of 80 objective-weight policies.

Perfect near-term price information therefore addresses only part of the control problem. The foresight arm supplied exact four-hour and 24-hour price averages, while the hindsight optimizer knew the full week. Service outcomes still depended on the resulting policy and its management of delayed obligations. The secondary results should be read with those information conditions and service outcomes attached.

The battery-capacity evidence covers 20 and 40 MWh. In the 40-MWh PPO group, two seeds produced nine service violations across four weeks. A larger store changes the available dispatch choices, while the learned policy must still manage those choices consistently. The present capacity comparison leaves the combined effects of charging power, efficiency, degradation, and controller design unresolved. The accounting replay, with zero discrepancies in all 27 checked quantities, supports the numerical consistency of these comparisons; the physical adequacy of the simulation remains a separate validation requirement.

5.4 Effect of historical price information on a fixed policy

The fourth question isolates a change that could be implemented without altering the policy's thresholds: replace current-price persistence with summaries derived from the previous week. Both versions met the service requirement throughout the evaluation. The resulting cost reduction was USD 37,691–106,254 per configuration over the 103 weeks, corresponding to 5.075%–9.162% of the persistence-to-hindsight opportunity. The opportunity-weighted effect across configurations was 7.414%.

The unchanged thresholds make the input change a specific, interpretable source of the measured difference. Its effect also varies over time. Each configuration improved in 33 weeks; the four battery configurations worsened in three other weeks. Reporting that pattern alongside the aggregate reduction shows the circumstances in which the changed summary helped or hurt. The USD 707,694.44 panel sum combines alternative facility designs, so the configuration-specific reductions are the relevant amounts for interpreting a single simulated design.

The remaining opportunity is substantial relative to the share captured by these historical summaries. Better forecasts and rolling-horizon optimization are reasonable subjects for a subsequent, separately specified comparison. Their value would need to be measured under the same service requirements and accounting rules. The current experiment establishes the descriptive effect of one causal input change on one fixed threshold policy.

5.5 Recurrent control and training duration

The fifth question concerns whether history conditioning or additional training changed service qualification. RecurrentPPO met the criterion in two of 20 policies, compared with nine of 20 companion feedforward PPO policies. It failed the all-five-seed requirement in every tested recurrent-control configuration. The companion comparison improves alignment of the training settings, but the recurrent network has about 29 times as many trainable parameters. The result applies to these two recipes, with memory and model capacity varying together.

Training duration produced a mixed pattern across the 12 planned comparisons: one compliance recovery, nine persistent failures, and two regressions. PPO with all five energy sources qualified at 2M and failed again at 5M. Its qualifying 2M result still cost 35.452% more than the service-compliant hindsight reference, with a 95% interval of 30.886%–41.928%. PPO with grid, solar, wind, and battery lost its 1M qualification at both later snapshots. These trajectories show that service qualification at one training budget can change at the next. They support checking the planned checkpoints individually and retaining the full seed population.

All six follow-up cost-comparison families were incomplete after the service screen. Increasing training interactions or adopting the tested recurrent recipe therefore failed to establish a qualified cost advantage within the specified comparisons. The tested budgets provide evidence about budget sensitivity; convergence and the general effectiveness of recurrent control require different evidence.

5.6 Use of the benchmark in controller assessment

The benchmark supports a staged assessment of candidate controllers. A candidate first has to satisfy the modeled service requirement across the declared training seeds. Cost comparisons then use complete evaluation episodes, common terminal adjustments, paired weeks, and the predeclared statistical procedure. A same-model hindsight reference identifies remaining cost opportunity under perfect information. Reporting these outcomes separately allows a reader to distinguish a service-qualified policy, a descriptive cost advantage, and a supported statistical claim.

For operational use, the service requirement and physical model would need to represent the facility's actual workload obligations, equipment limits, reserve needs, and control authority. A model-based evaluation could then screen candidates before a monitored trial. The present study supplies an evaluation framework and recorded evidence from one simulated setting. Its business value would depend on whether a candidate subsequently delivers measurable savings while satisfying the site's operational requirements.

6. Conclusion

The Data-Centre Energy-Control Claim Qualification Benchmark evaluates the conditions required to substantiate a controller cost advantage. Its primary result is that service compliance across independently trained policies sharply limits which comparisons qualify: 136 of 240 learned policies passed individually, while four of 48 algorithm–configuration combinations passed in all five seeds. This left the predeclared primary cost-comparison family incomplete and prevented a Holm-corrected superiority claim against the rule-based baseline.

The four qualified learned combinations retained hindsight gaps of 14.302%–24.815% of controller cost. Supplying a fixed threshold policy with historical weekly price summaries reduced cost by USD 37,691–106,254 per configuration while maintaining service compliance throughout the evaluation. That input change captured 7.414% of the opportunity-weighted persistence-to-hindsight gap. Recurrent control and additional training produced different service outcomes across seeds and budgets: 50 of 100 follow-up policies qualified individually, and the 12 primary budget comparisons yielded one recovery, nine persistent failures, and two regressions. All six follow-up cost-comparison families remained incomplete.

These findings establish a practical role for qualification-based reporting: make service outcomes, seed variability, complete accounting, information conditions, and statistical support visible before presenting a controller advantage. They identify specific questions for further controller development, including how forecast quality, dispatch planning, and training design affect service-qualified costs. Prospective confirmation, broader physical validation, and monitored facility trials are the next evidence needed to assess transfer beyond this simulation.

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